Understanding Dataset Difficulty with V-Usable Information

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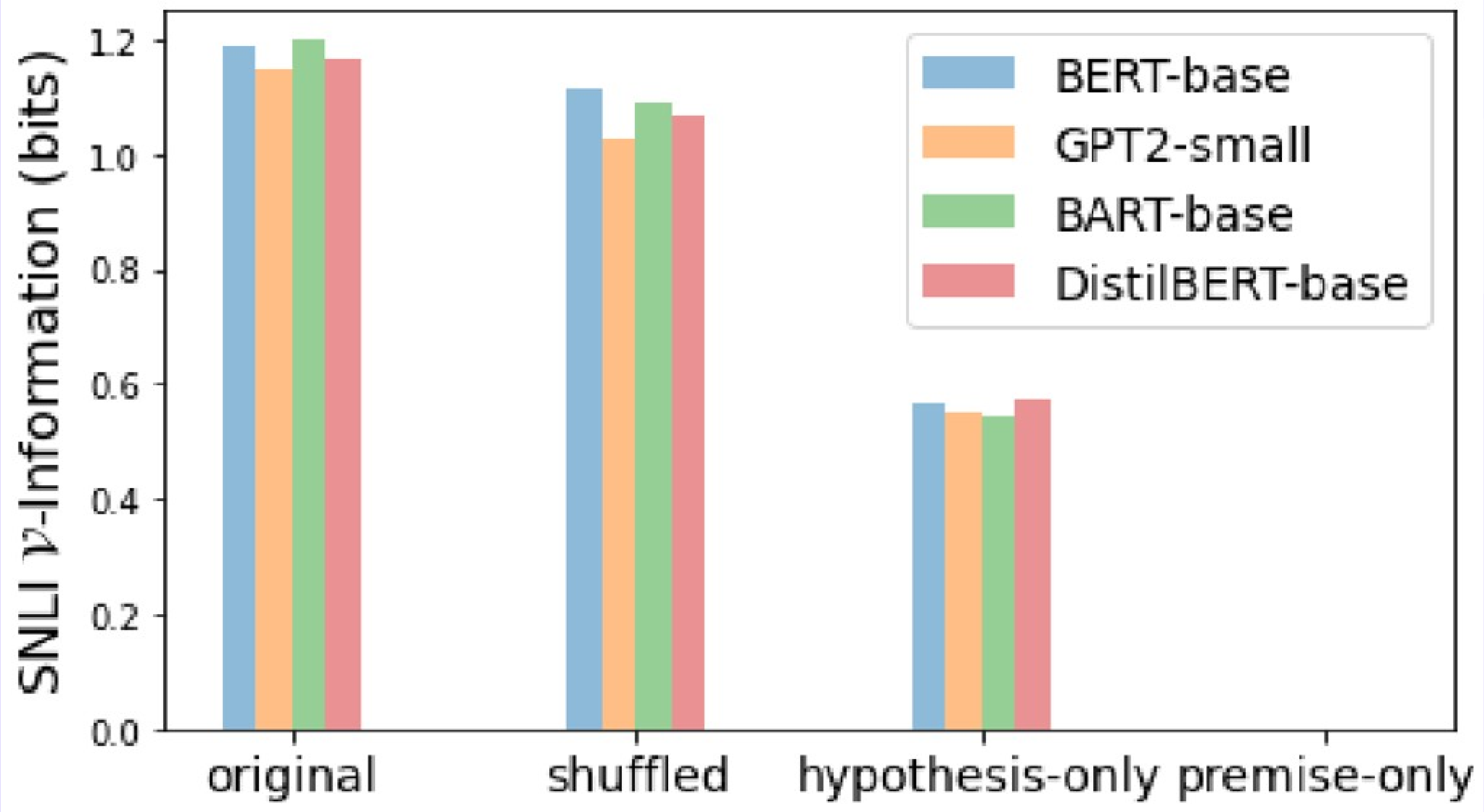
01 Our Method at a Glance (Visual)



- We show the pipeline for I_V and PVI.
- We train prior-only and input-using models in V.

02 We Recast Dataset Difficulty with I_V and PVI

- We define dataset difficulty as lack of V-usable information.
- We introduce PVI to measure instance-level difficulty within distributions.
- We compare datasets, slices, and instances for a fixed model family.
- We attribute difficulty to inputs via transformations and attributions.
- We interpret higher I_V as easier data for the chosen V.

03 Motivation: From I_V to Instance-Level PVI

- We need instance-level signals beyond dataset-level I_V.
- We use PVI to see which items are easy or hard.
- We summarize slice difficulty with mean PVI across classes or domains.
- We surface mislabels and spurious priors with low PVI.
- We track how attributes shift usable signal with transformations.

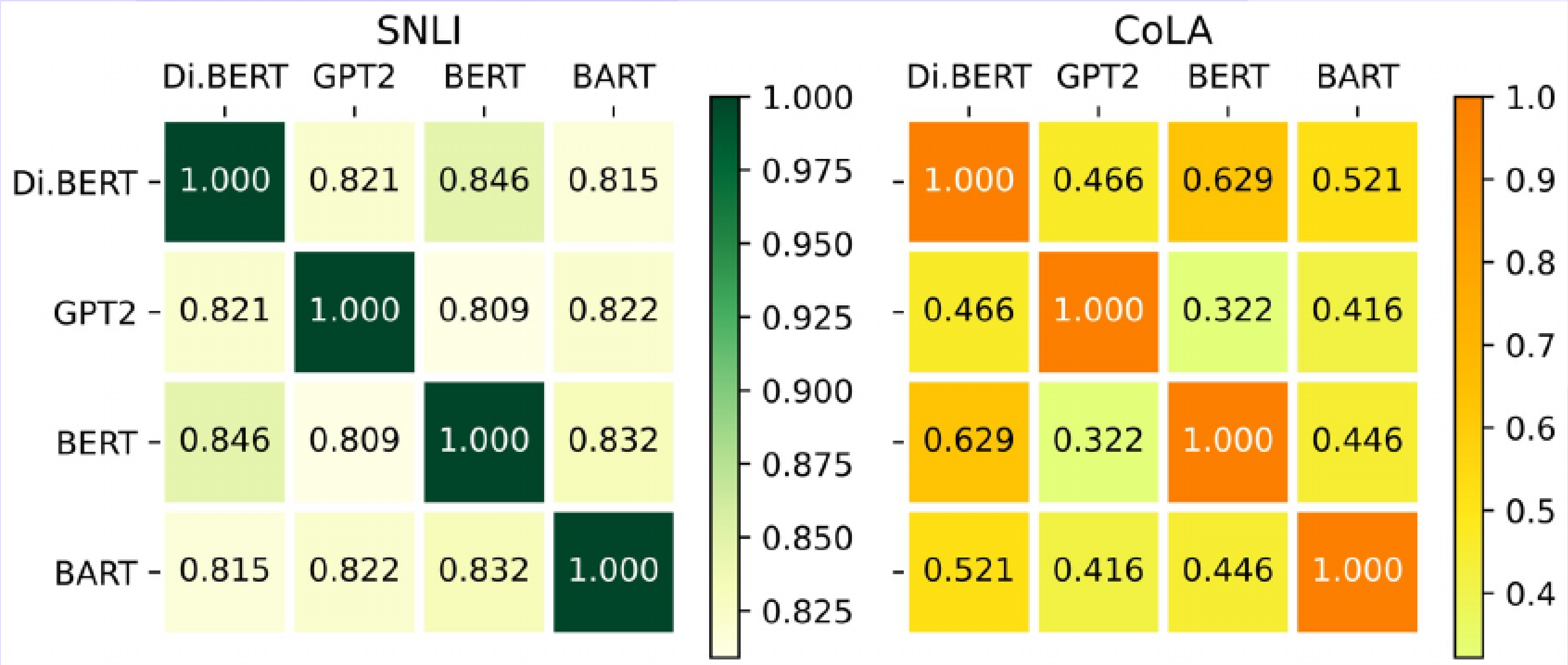
04 V-Information: Intuition, Estimation, and Assumptions

- We measure V-usable information for a fixed family.
- We estimate on held-out cross-entropy, with and without X.
- We assume i.i.d. splits and sufficient data.
- We compare models, datasets, and inputs under fixed Y.
- We prefer simpler V for tighter bounds, stable estimates.
- We show I_V/accuracy vs epochs.

05 We Validate I_V vs. Accuracy and Overfitting

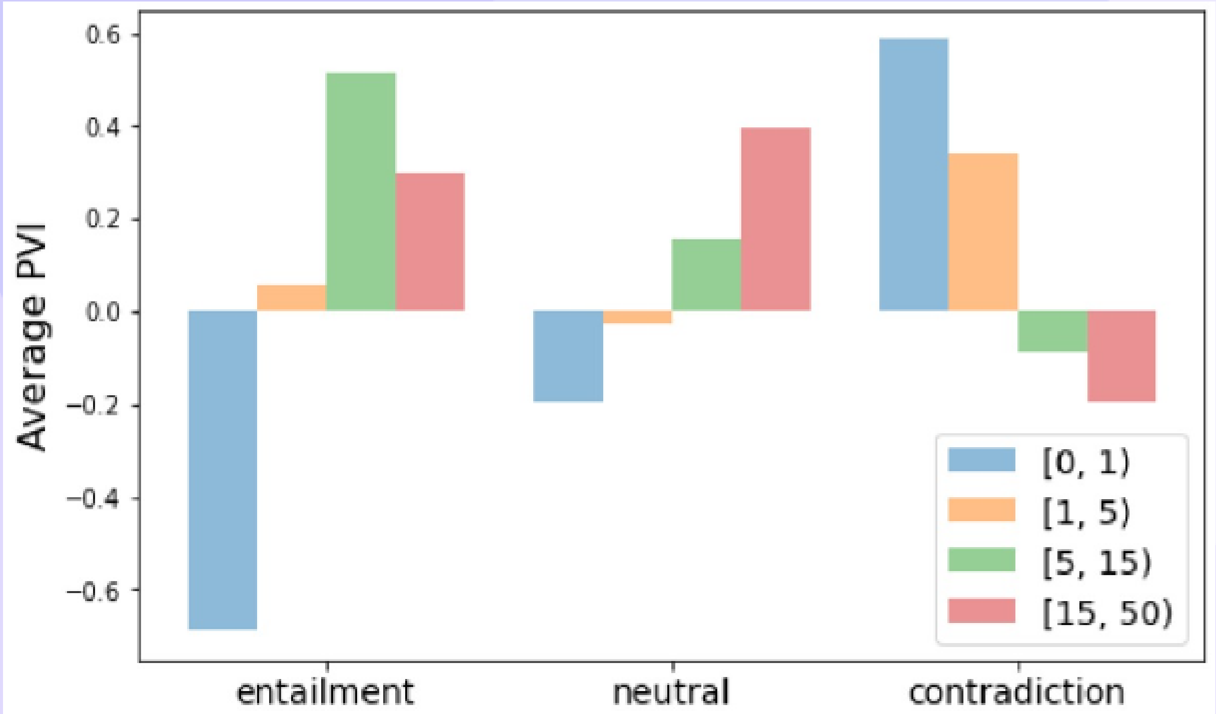
- We find I_V tracks accuracy across NLI models.
- We see overfitting as I_V falls before accuracy.
- We compare datasets: SNLI > MultiNLI; CoLA hardest.
- We interpret higher I_V as higher held-out certainty.
- We use low I_V to flag harder data, weak cues.
- Top: I_V; Bottom: accuracy vs epochs.

06 Pointwise V-Information (PVI): Definition and Uses



- We define $\mathrm{PVI}(x \rightarrow y) = -\log g_{\varnothing}(y) + \log g'_x(y)$.
- We recover I_V by averaging PVI over the dataset.
- We observe high cross-model and cross-seed correlations on easy data.
- We use low PVI to flag mislabels and distribution issues.
- We compare only within the same inputs, labels, and V.

07 Attribute Probing and Dataset Slicing with Mean PVI



- We find shuffling retains most usable NLI information.
- We show hypothesis-only remains informative, revealing artefacts.
- We compare classes and slices using mean PVI.
- We see overlap helps entailment more than others.
- We use slice means to guide curation and monitoring.
- Bars: mean PVI by class and attribute.

08

Hate Speech: Token-Level Artefacts and Conditional V-Information

Grammatical	Ungrammatical
will (0.267)	book (2.737)
John (0.168)	is (2.659)
. (0.006)	was (2.312)
and (-0.039)	of (2.308)
in (-0.05)	to (1.972)

- We reveal strong lexical biases via leave-one-out omissions.
- We measure offensive words carrying most BERT-usable signal.
- We show ungrammaticality cues cluster on auxiliaries and prepositions.

- We isolate information from offensive tokens with conditional I_V.
- We warn lexical shortcuts can cause harm and miss nuance.

Thank You

Questions & Discussion